# ReadWise AI — Pediatric Dyslexia Risk Screening Report

Generated on October 21, 2026 at 07:39 PM | Standard Multimodal Protocol

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|---|---|---|
| **Child Name:** Arun | **Age:** 9 years | **Grade:** Grade 4 |
| **Assessment Task:** T4_Meaningful_Text | **Primary Language:** English | **Session ID:** #1009 |

## AI SCREENING ESTIMATE: HIGH DYSLEXIA RISK INDICATOR

Estimated Risk Probability: **78.0%** | Classification Confidence: **89.0%**

***IMPORTANT NOTICE:*** *This screening result is an educational early-warning assessment designed to triage support. It does NOT constitute a formal clinical or medical diagnosis of dyslexia.*

## 1. Multimodal Behavioral Analytics Summary

| Modality | Observed Metric | Measured Value | Typical Cohort Reference | Status |
|---|---|---|---|---|
| Eye-Tracking | Gaze Regressions | 32 events | < 18 events | Elevated Backtracking |
| Eye-Tracking | Dwell Time | 120.0 s | < 80.0 s | Extended Fixation |
| Eye-Tracking | Progress/Regress Ratio | 3.8 | > 7.0 | Reduced Fluency |
| Speech | Hesitation Rate (>500ms) | 4.8 /min | < 2.0 /min | Frequent Pauses |
| Speech | Oral Reading WPM | 64.0 WPM | 85 - 110 WPM | Below Grade Level |
| Cognitive | Phonological Score | 52 / 100 | > 80 / 100 | Support Recommended |
| Cognitive | Working Memory | 58 / 100 | > 80 / 100 | Mild Delay |

## 2. Explainable AI (XAI) Transparent Attribution

Using SHAP (Shapley Additive exPlanations), the relative impact of each observed factor on the screening outcome is transparently quantified below:

| Feature Factor | SHAP Impact | Clinical/Educational Explanation |
|---|---|---|
| Speech Hesitation Rate | +0.082 | Long inter-word silent gaps indicate difficulty retrieving phonological word representations. |
| Eye-Gaze Regressions | +0.065 | Involuntary saccadic jumps backward in the line indicate decoding breakdown. |
| Low Phonological Score | +0.058 | Challenges in blending individual phonemes into whole words. |
| Extended Dwell Duration | +0.034 | High total fixation time across syllables reflects cognitive reading fatigue. |

## 3. Recommended Personalized Learning Plan

**Recommended Daily Practice:** 25 minutes | **Primary Goal:** Strengthen phonics decoding and reduce visual reading regressions.

| Activity Title | Target Domain | Recommended Daily Time | Priority |
|---|---|---|---|
| Phonics Sound-Blending Quest | Phonological Awareness | 12 minutes | Critical |
| Visual Tracking & Anti-Regression Maze | Oculomotor Reading Flow | 8 minutes | High |
| Rapid Sight-Word Recognition | Reading Fluency | 10 minutes | High |

***Reviewer Guidelines:*** *This document should be reviewed in consultation with an educational specialist, reading coach, or speech-language pathologist. Continue daily progress tracking within the ReadWise AI application.*

*ReadWise AI Research Project — Karunya Institute of Technology & Sciences*